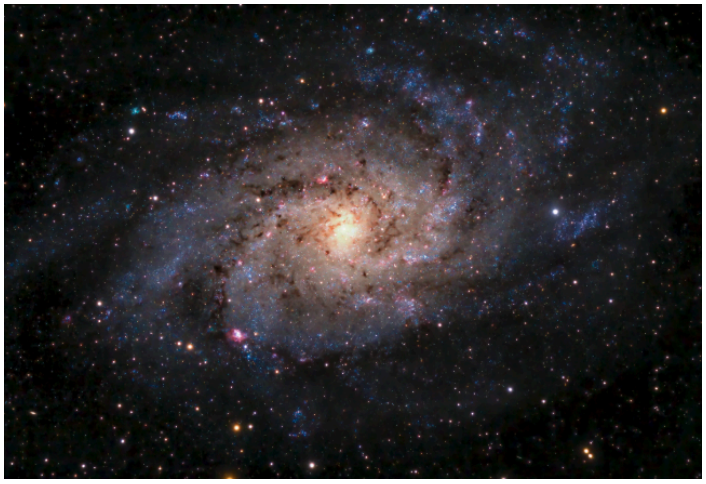
Astronomical Messier-Object Image Identifier

This project explores training a model to recognize astronomical objects in the Messier catalog and classify new telescope images accordingly. While many classifiers exist for modern deep-sky objects, this project focuses on a more constrained, interpretable mapping of images to specific Messier categories—galaxies, clusters, and nebulae. The model would learn from historical catalog images as well as modern CCD/CMOS amateur astronomy datasets to generalize across exposure length, noise, and stacking differences. This approach leads to a useful educational tool for amateur astronomers or automated tagging for astrophotography.

Data Needed & Methods

- Images of all 110 Messier objects from public datasets (e.g., NASA, ESA, DeepSkyStacker communities).
- Preprocessing: noise reduction, histogram equalization, cropping, and resizing.
- Possible feature extraction: star-field density, texture patterns, central brightness, or CNN-extracted embeddings.
- Multiclass classifier: CNN, transfer learning from ResNet/VGG, or even classical ML on extracted features.
- Evaluation using top-1 accuracy and confusion matrices to examine class-confusion (e.g., similar nebulae).
- Bonus extension: attempt identification on noisy or partially occluded images.



M-33 Triangulum Galaxy

Definition Clarifiers:

UV/IR (Ultraviolet / Infrared)

UV and IR refer to portions of the electromagnetic spectrum outside visible light.

In imaging contexts, **UV** wavelengths can reveal security markers or fluorescent features, while

IR captures heat signatures or ink patterns invisible to normal cameras.

Some counterfeit detection and forensic imaging systems use UV/IR filters to expose hidden details in currency, artwork, or documents.